

SARCATHON 2026

SEMICON HACKATHON

DESIGN THE BRIDGE. CONNECT THE FUTURE.

Can you build the hardware that connects high-performance SoC systems to PCIe? Design an AXI4 ↔ PCIe 6 interface, verify it against real-world workloads, and take your design from RTL all the way to GDSII.

THE CHALLENGE

AXI4 is widely used for communication inside SoCs and accelerator systems, while **PCIe** is used to connect high-performance hardware to a host system. Your task is to develop the RTL hardware that enables these interfaces to communicate.

The challenge targets applications such as **network packet processing** and **machine-learning inference accelerators**, where large amounts of data move between accelerator hardware and host memory.

STAGE 1 — QUALIFIER DESIGN & VERIFY THE RTL

Build the core AXI4-to-PCIe interface at the RTL level and demonstrate that the design works correctly through simulation.

Your tasks:

- Understand the given architecture and specifications
- Develop the required Verilog/SystemVerilog RTL
- Build a functional testbench
- Verify key read/write transactions and reset/error behaviour
- Demonstrate simulation results

Stage 1 Deliverables: RTL + Testbench + Simulation Results + Short Design Report

STAGE 2 — GRAND FINALE OPTIMIZE. IMPLEMENT. TAPE OUT.

Take your working RTL to the next level: improve the design, validate it against application workloads, and complete the digital implementation flow.

Your tasks:

- Refine and optimize the RTL for performance, area and reliability
- Perform exhaustive functional verification
- Test network packet-processing and ML-inference workloads
- Synthesize the RTL and analyze implementation results
- Complete physical design using OpenROAD/OpenLane
- Generate the final GDSII

Bonus: FPGA-tested AXI4-to-PCIe IP

FROM RTL TO GDSII

Specification → Architecture → RTL → Simulation → Verification → Synthesis → Physical Design
→ GDSII

TOOLS & RESOURCES

HDL / Verification	Simulation / FPGA	Physical Design
Verilog / SystemVerilog / SVA Python	Icarus Verilog Xilinx Vivado	OpenROAD / OpenLane Open-source PDKs

ARE YOU READY TO BUILD THE HARDWARE BEHIND THE CONNECTION?

Compete • Design • Verify • Implement • Innovate